

# Onyx — GPIOKit reference

Onyx · multi-process OS on Raspberry Pi 4

The reference of **GPIOKit** (`SD:/lib/gpiokit.so`, `user/Kits/gpiokit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: [The Kits](#).

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## What it is

GPIOKit is the Raspberry Pi's 40-pin header: a pin's mode and level, PWM, a servo, edges queued with their time, the I2C bus and SPI, the header's own table — and a simulated board (an SSD1306 display and a BME280 sensor on its I2C bus) for a PC or a system without the hardware. The levels are 3.3 V.

|         |                                                                                                                          |
|---------|--------------------------------------------------------------------------------------------------------------------------|
|         |                                                                                                                          |
| Include | <code>#include "gpiokit/gpiokit.h"</code>                                                                                |
| Link    | <code>lib/gpiokit.imp.a</code> (C++) or <code>lib/gpiokit.imp_c.a</code> (C)                                             |
| Library | <code>SD:/lib/gpiokit.so</code> — 34 entries in its table<br>( <code>user/Kits/gpiokit/gpiokit.abi</code> , append-only) |
| Sources | <code>user/Kits/gpiokit/</code>                                                                                          |

## Using it

The Raspberry Pi's header from a program: a pin's mode, its level, PWM, edges, the I2C bus and SPI. Pins are **BCM GPIO numbers** (GPIO 17 is the header's pin 11: `gk_header_pin`). **The levels are 3.3 V**: never wire 5 V to a pin; at most about 16 mA a pin (an LED through a 330  $\Omega$  resistor). A pin is the program's from its first use until it gives it back (`gk_mode (pin, GK_FREE)`, `gk_release`) or ends — the system then makes it an input again, after a crash too. The system's pins (GPIO 14 / 15, the serial console; 0 / 1, the HAT EEPROM) and a pin another program has are refused: every call returns  $\geq 0$ , or a negative `GK_E*` that `gk_error` says in words.

GPIOKit is the one kit besides AppKit that calls the kernel itself (kapi v92 `gpio_ctl`; the user's exception, docs/03 §5.10); it is shipped with the kernel in the package `onyx`.

**An LED, a button (C):**

```
#include "gpiokit/gpiokit.h"
```

```

gk_mode (17, GK_OUT); // the LED: GPIO 17 -> 330 ohm
-> LED -> GND
gk_mode (27, GK_IN_PULLUP); // the button: GPIO 27 ->
button -> GND (reads 0 pressed)
gk_edges (27, GK_FALLING); // its presses queued, with
their time
for (;;)
{
    gk_event e[8];
    int n = gk_events (e, 8, 1000); // waits up to 1 s for the
first
    for (int i = 0; i < n; i++) gk_toggle (17); // each press turns the LED
over
}

```

**PWM and a servo** (GPIO 12, 13, 18, 19; 12 / 18 and 13 / 19 share a channel):

```

gk_pwm (18, 1000, 2500); // 1 kHz, 25 % high (duty in
1/10000)
gk_servo (18, 1500); // a servo: 50 Hz, a 1.5 ms
pulse (the middle)

```

**I2C** (GPIO 2 SDA, GPIO 3 SCL) — a BME280's id, a scan:

```

if (gk_i2c_open (0) == 0) // 0: 100 kHz
{
    unsigned char map[16];
    int n = gk_i2c_scan (map); // bit a of map: something
answers at a
    if (gk_i2c_reg_read (0x76, 0xD0) == 0x60) ax_putln ("a BME280 at 0x76");
    gk_i2c_close ();
}

```

**SPI 0** (GPIO 8 CE0, 7 CE1, 9 MISO, 10 MOSI, 11 SCLK): `gk_spi_open (1000000, 0)`, then `gk_spi_transfer (0, tx, rx, n)`.

**C++** has small classes that give the pin back by themselves:

```

gpiokit::Pin led (17, GK_OUT);
if (!led.ok ()) ax_putln (gk_error (led.status ()));
led.write (1);
gpiokit::I2CDevice bme (0x76);
int id = bme.reg (0xD0);

```

**No hardware?** GPIOKit has a **simulator** — a board in memory: the inputs set by `gk_sim_input`, the PWM's waves, an I2C bus with an SSD1306 display (0x3C: `gk_sim_display` gives its pixels) and a BME280 sensor (0x76), SPI looped back. `gk_sim (1)` switches to it; it runs by itself where the system has no GPIO, and on a PC. GPIO Lab (docs/04) is built on GPIOKit; BASIC has statements for it (docs/04 §13 *GPIO*).

## Index

Everything the headers declare, in their order — the details are in each header's part below.

| Name                     | What it does                                         | Header                 |
|--------------------------|------------------------------------------------------|------------------------|
| <code>gk_pin_info</code> | A pin as the system sees it ( <code>gk_info</code> ) | <code>gpiokit.h</code> |
| <code>gk_event</code>    | An edge ( <code>gk_events</code> )                   | <code>gpiokit.h</code> |

| Name                      | What it does                                                                               | Header                 |
|---------------------------|--------------------------------------------------------------------------------------------|------------------------|
| <code>gk_available</code> | What drives the pins                                                                       | <code>gpiokit.h</code> |
| <code>gk_sim</code>       | The simulator on (1                                                                        | <code>gpiokit.h</code> |
| <code>gk_error</code>     | A negative code's text ("the system uses this pin").                                       | <code>gpiokit.h</code> |
| <code>gk_release</code>   | Every pin, bus and edge of the program's given back (its end does it too) → 0.             | <code>gpiokit.h</code> |
| <code>gk_now_us</code>    | The clock of the edges' times, now (microseconds                                           | <code>gpiokit.h</code> |
| <code>gk_mode</code>      | The pin set to a mode (GK_FREE ...                                                         | <code>gpiokit.h</code> |
| <code>gk_write</code>     | An output of the program's set high (1) or low (0) → 0 / GK_E*.                            | <code>gpiokit.h</code> |
| <code>gk_toggle</code>    | An output turned over → its new level / GK_E*.                                             | <code>gpiokit.h</code> |
| <code>gk_read</code>      | A pin's level now → 0 / 1 / GK_EINVAL (any pin of the header may be read, whoever has it). | <code>gpiokit.h</code> |
| <code>gk_read_all</code>  | Every level at once                                                                        | <code>gpiokit.h</code> |
| <code>gk_info</code>      | The pins as the system sees them, GPIO 0 first, up to max (GK_PINS                         | <code>gpiokit.h</code> |
| <code>gk_mode_name</code> | A mode's name ("input", "pull-up", "output", "PWM", "I2C", "ALT3"...).                     | <code>gpiokit.h</code> |
| <code>gk_pwm</code>       | GPIO 12 / 13 / 18 / 19 made a square wave of freq_hz (1 ..                                 | <code>gpiokit.h</code> |
| <code>gk_servo</code>     | A hobby servo on a PWM pin                                                                 | <code>gpiokit.h</code> |
| <code>gk_edges</code>     | An input of the program's                                                                  | <code>gpiokit.h</code> |
| <code>gk_events</code>    | The edges that came, up to max, waiting up to wait_ms (0 ..                                | <code>gpiokit.h</code> |
| <code>gk_i2c_open</code>  | The I2C bus 1 (GPIO 2 SDA, GPIO 3 SCL                                                      | <code>gpiokit.h</code> |
| <code>gk_i2c_close</code> | The bus given back → 0.                                                                    | <code>gpiokit.h</code> |

| Name                           | What it does                                                                                      | Header                 |
|--------------------------------|---------------------------------------------------------------------------------------------------|------------------------|
| <code>gk_i2c_write</code>      | n bytes written to the device at addr (7 bits)                                                    | <code>gpiokit.h</code> |
| <code>gk_i2c_read</code>       | n bytes read from the device → n / GK_EIO.                                                        | <code>gpiokit.h</code> |
| <code>gk_i2c_write_read</code> | wn bytes written, then rn read (a repeated start between)                                         | <code>gpiokit.h</code> |
| <code>gk_i2c_reg_read</code>   | A register of 8 bits read → its value (0..255) / GK_E*.                                           | <code>gpiokit.h</code> |
| <code>gk_i2c_reg_write</code>  | A register of 8 bits written → 0 / GK_E*.                                                         | <code>gpiokit.h</code> |
| <code>gk_i2c_scan</code>       | Every address 0x08..0x77 tried                                                                    | <code>gpiokit.h</code> |
| <code>gk_i2c_guess</code>      | What usually answers at an I2C address ("SSD1306 display", "BME280 / BMP280 sensor"...), or "".   | <code>gpiokit.h</code> |
| <code>gk_spi_open</code>       | SPI 0 the program's (GPIO 8 CE0, 7 CE1, 9 MISO, 10 MOSI, 11 SCLK), at clock_hz (0                 | <code>gpiokit.h</code> |
| <code>gk_spi_transfer</code>   | n bytes sent while n come back, chip select cs (0                                                 | <code>gpiokit.h</code> |
| <code>gk_spi_close</code>      | SPI 0 given back → 0.                                                                             | <code>gpiokit.h</code> |
| <code>gk_header_gpio</code>    | The 40 pins as printed on the board, pin 1 (3V3, top left with the board's USB ports down) to 40. | <code>gpiokit.h</code> |
| <code>gk_header_pin</code>     | A GPIO's header pin → 1..40, or -1.                                                               | <code>gpiokit.h</code> |
| <code>gk_header_label</code>   | A header pin's label                                                                              | <code>gpiokit.h</code> |
| <code>gk_gpio_function</code>  | What else a GPIO is, if anything                                                                  | <code>gpiokit.h</code> |
| <code>gk_sim_input</code>      | (gk_sim (1) first, or a PC build.) What drives a simulated input from outside                     | <code>gpiokit.h</code> |
| <code>gk_sim_display</code>    | What the simulated SSD1306 shows                                                                  | <code>gpiokit.h</code> |
| <code>Pin</code>               | A pin of the program's                                                                            | <code>gpiokit.h</code> |
| <code>Pwm</code>               | A PWM output (GPIO 12, 13, 18 or 19).                                                             | <code>gpiokit.h</code> |

| Name      | What it does                                                                   | Header    |
|-----------|--------------------------------------------------------------------------------|-----------|
| I2CDevice | A device on the I2C bus (the bus opened by the first one made, if it was not). | gpiokit.h |

## gpiokit/gpiokit.h

gpiokit.h -- GPIOKit, the Raspberry Pi's 40-pin header for the programs (SD:/lib/gpiokit.so; docs/06-KITS-GUIDE.md "GPIOKit", docs/19-GPIOKIT.md).

```

pins          a pin's mode (input, pulled up / down, output, an alternate function),
its level
              written and read, every level at once (gk_mode, gk_write, gk_read,
gk_read_all)
PWM          a square wave on GPIO 12 / 13 / 18 / 19: its frequency and its duty; a
servo's pulse
              (gk_pwm, gk_servo)
edges        a pin's rising / falling edges queued, with their time, and taken
(gk_edges, gk_events)
I2C          the bus on GPIO 2 (SDA) / 3 (SCL): transfers, a device's registers, a
scan (gk_i2c_*)
SPI          SPI 0 on GPIO 7..11: full-duplex transfers (gk_spi_*)
the header   the 40 pins as printed on the board: their numbers, names, functions
(gk_header_*)
simulator    a board in memory -- inputs set by hand, the PWM's waves, an I2C bus
with an SSD1306
              display and a BME280 sensor, SPI looped back -- for a PC, or an Onyx
without the
              hardware (gk_sim, gk_sim_input)

```

Pins are BCM GPIO numbers (GPIO 17 is the header's pin 11). The levels are 3.3 V: never wire 5 V to a pin. A pin is the program's from its first use until gk\_mode (pin, GK\_FREE), gk\_release or the program's end -- then the system makes it an input again. Pins the system uses (the serial console's GPIO 14 / 15, the HAT EEPROM's 0 / 1) are refused, and so is a pin another program has.

Every call that can fail returns a negative GK\_E\* code (gk\_error says it in words); >= 0 is success. A C++ program also has small classes over these (namespace gpiokit: Pin, Pwm, I2CDevice).

Link lib/gpiokit.imp.a (C++) or lib/gpiokit.imp\_c.a (C). The interface is append-only (gpiokit.abi). GPIOKit is the one kit besides AppKit that calls the kernel's table itself (kapi gpio\_ctl, v92; the user's exception, docs/03 §5.10): it is shipped and rebuilt with the kernel, in the package onyx.

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```
#define GK_PINS      28      // GPIO 0..27: the header's
```

## modes, edges, errors

(the same numbers as the kernel's KAPI\_GPIO\_M\_\*)

```
#define GK_FREE      0      // nobody's: an input, no pull (gk_mode gives the pin
back)
#define GK_IN        1      // an input, floating
#define GK_IN_PULLUP  2      // an input pulled up (a button to ground reads 0
when pressed)
#define GK_IN_PULLDOWN 3      // an input pulled down
#define GK_OUT        4      // an output (starts low)
#define GK_PWM        5      // a PWM output (gk_pwm sets it)
#define GK_I2C        6      // a pin of the I2C bus (gk_i2c_open)
#define GK_SPI        7      // a pin of SPI 0 (gk_spi_open)
#define GK_ALT0       8      // GK_ALT0 + n: alternate function n (0..5) chosen by
hand

#define GK_RISING     1      // gk_edges: the edges wanted
#define GK_FALLING    2
#define GK_BOTH       3

#define GK_EPERM      (-1)    // the system's pin, or not the program's
#define GK_EIO        (-5)    // no answer on the bus
#define GK_ENOMEM      (-12)
#define GK_EFAULT      (-14)
#define GK_EBUSY       (-16)   // another program has it
#define GK_ENODEV      (-19)   // no GPIO here (a system older than kapi 92): gk_sim
(1) simulates
#define GK_EINVAL      (-22)   // a wrong pin, mode, frequency, duty...
```

gk\_pin\_info.flags

```
#define GK_F_RESERVED  1      // the system's: never given (reason says whose)
#define GK_F_PWM       2      // can do PWM (12, 13, 18, 19)
#define GK_F_EDGES     4      // its edges are queued for its owner
```

A pin as the system sees it (gk\_info): the same layout as the kernel's struct kapi\_gpio\_pin.

```
typedef struct gk_pin_info
{
    unsigned char pin;      // GPIO number
    unsigned char mode;     // GK_FREE ... GK_ALT0 + 5
    unsigned char level;    // 0 / 1 now
    unsigned char flags;    // GK_F_*
    unsigned owner;         // the program (its pid) that has it, 0: none
    unsigned pwm_freq;      // Hz, in PWM
    unsigned pwm_duty;      // 1/10000, in PWM
    char reason[24];        // why it is reserved ("serial console")
} gk_pin_info;
```

An edge (gk\_events): the same layout as the kernel's struct kapi\_gpio\_event.

```
typedef struct gk_event
{
    unsigned char pin;
    unsigned char edge;    // GK_RISING / GK_FALLING
    unsigned char level;   // the level just after it
    unsigned char lost;    // 1: edges were dropped before this one (the queue was
full)
    unsigned reserved;
    unsigned long long us;    // when, in microseconds (gk_now_us's clock)
} gk_event;
```

## the kit

What drives the pins: 1 the Pi's header, 2 the simulator, 0 nothing (gk\_sim (1) gives the simulator).

```
int gk_available (void);
```

The simulator on (1: every call after goes to a board in memory) or off (0: the real header, if there is one) → what drives the pins now (as gk\_available). A PC build of GPIOKit is always simulated.

```
int gk_sim (int on);
```

A negative code's text ("the system uses this pin").

```
const char *gk_error (int code);
```

Every pin, bus and edge of the program's given back (its end does it too) → 0.

```
int gk_release (void);
```

The clock of the edges' times, now (microseconds; it only goes forward).

```
unsigned long long gk_now_us (void);
```

## pins

The pin set to a mode (GK\_FREE ... GK\_OUT, GK\_ALT0 + n) → 0 / GK\_E\*.

```
int gk_mode (int pin, int mode);
```

An output of the program's set high (1) or low (0) → 0 / GK\_E\*.

```
int gk_write (int pin, int level);
```

An output turned over → its new level / GK\_E\*.

```
int gk_toggle (int pin);
```

A pin's level now → 0 / 1 / GK\_EINVAL (any pin of the header may be read, whoever has it).

```
int gk_read (int pin);
```

Every level at once: bit n = GPIO n.

```
unsigned gk_read_all (void);
```

The pins as the system sees them, GPIO 0 first, up to max (GK\_PINS: all) → how many / GK\_E\*.

```
int gk_info (gk_pin_info *out, int max);
```

A mode's name ("input", "pull-up", "output", "PWM", "I2C", "ALT3"...).

```
const char *gk_mode_name (int mode);
```

## PWM

GPIO 12 / 13 / 18 / 19 made a square wave of freq\_hz (1 .. 1 000 000) high for duty / 10000 of the time (0 .. 10000: 2500 = 25 %) → 0 / GK\_E\*. 12 and 18 share a channel, 13 and 19 the other: one of each pair. The pin stays in PWM until gk\_mode gives it another mode.

```
int gk_pwm (int pin, int freq_hz, int duty);
```

A hobby servo on a PWM pin: 50 Hz, a pulse of pulse\_us (500 .. 2500; 1500: the middle) → 0 / GK\_E\*.

```
int gk_servo (int pin, int pulse_us);
```

## edges

An input of the program's: its edges (GK\_RISING | GK\_FALLING; 0: none) queued for it → 0 / GK\_E\*. The queue holds 256 edges; a button bounces: a few edges a press.

```
int gk_edges (int pin, int edges);
```

The edges that came, up to max, waiting up to wait\_ms (0 .. 1000) for the first → how many / GK\_E\*.

```
int gk_events (gk_event *out, int max, int wait_ms);
```

## I2C

The I2C bus 1 (GPIO 2 SDA, GPIO 3 SCL; the board has its pull-ups) the program's, at clock\_hz (0: 100 kHz; 400000 for fast devices) → 0 / GK\_E\*.

```
int gk_i2c_open (int clock_hz);
```

The bus given back → 0.

```
int gk_i2c_close (void);
```

n bytes written to the device at addr (7 bits: 0x3C) → n / GK\_EIO (no answer).

```
int gk_i2c_write (int addr, const void *data, int n);
```

n bytes read from the device → n / GK\_EIO.

```
int gk_i2c_read (int addr, void *data, int n);
```

wn bytes written, then rn read (a repeated start between: a register's address, then its bytes) → rn / GK\_E\*.

```
int gk_i2c_write_read (int addr, const void *wr, int wn, void *rd, int rn);
```

A register of 8 bits read → its value (0..255) / GK\_E\*.

```
int gk_i2c_reg_read (int addr, int reg);
```

A register of 8 bits written → 0 / GK\_E\*.

```
int gk_i2c_reg_write (int addr, int reg, int value);
```

Every address 0x08..0x77 tried: bit a of map (16 bytes: map[a / 8] >> (a % 8)) set where a device answered → how many answered / GK\_E\*.

```
int gk_i2c_scan (unsigned char map[16]);
```

What usually answers at an I2C address ("SSD1306 display", "BME280 / BMP280 sensor"...), or "".

```
const char *gk_i2c_guess (int addr);
```

## SPI

SPI 0 the program's (GPIO 8 CEO, 7 CE1, 9 MISO, 10 MOSI, 11 SCLK), at clock\_hz (0: 1 MHz), mode 0..3 (CPOL << 1 | CPHA) → 0 / GK\_E\*. Again: the clock and the mode changed.

```
int gk_spi_open (int clock_hz, int mode);
```

n bytes sent while n come back, chip select cs (0: CEO, 1: CE1); tx 0: zeros sent, rx 0: dropped → n / GK\_E\*.

```
int gk_spi_transfer (int cs, const void *tx, void *rx, int n);
```

SPI 0 given back → 0.

```
int gk_spi_close (void);
```

## the header

The 40 pins as printed on the board, pin 1 (3V3, top left with the board's USB ports down) to 40. A header pin's GPIO → 0..27, or -1 (a power or ground pin).

```
int gk_header_gpio (int header_pin);
```

A GPIO's header pin → 1..40, or -1.

```
int gk_header_pin (int gpio);
```

A header pin's label: "3V3", "5V", "GND", "GPIO17"...

```
const char *gk_header_label (int header_pin);
```

What else a GPIO is, if anything: "SDA1", "SCL1", "TXD0", "PWM0", "SPI0 CEO", "ID\_SD"... or "".

```
const char *gk_gpio_function (int gpio);
```

## the simulator

(gk\_sim (1) first, or a PC build.) What drives a simulated input from outside: level 0 / 1, or -1 (nothing: the pin floats, or follows its pull) → 0. An edge is queued if the program asked for it.

```
int gk_sim_input (int pin, int level);
```

What the simulated SSD1306 shows: its 128 × 64 pixels, a byte for 8 rows (its own memory layout: page p, column c at p \* 128 + c) → out (1024 bytes) filled, 1 / 0 (the simulator is off).

```
int gk_sim_display (unsigned char out[1024]);
```

```
}
```

A pin of the program's: its mode set at its making, given back at its end.

```
class Pin
{
public:
```

```

Pin (int gpio, int mode) : gpio_ (gpio) { status_ = gk_mode (gpio, mode); }
~Pin () { if (status_ == 0) gk_mode (gpio_, GK_FREE); }
Pin (const Pin &) = delete;
Pin &operator= (const Pin &) = delete;
bool ok () const { return status_ == 0; }
int status () const { return status_; } // 0, or why it failed (gk_error)
int gpio () const { return gpio_; }
int read () const { return gk_read (gpio_); }
int write (int level) const { return gk_write (gpio_, level); }
int toggle () const { return gk_toggle (gpio_); }
int edges (int which) const { return gk_edges (gpio_, which); }
private:
    int gpio_, status_;
};

```

A PWM output (GPIO 12, 13, 18 or 19).

```

class Pwm
{
public:
    explicit Pwm (int gpio, int freq_hz = 1000, int duty = 0) : gpio_ (gpio), freq_
(freq_hz) { status_ = gk_pwm (gpio, freq_hz, duty); }
    ~Pwm () { if (status_ == 0) gk_mode (gpio_, GK_FREE); }
    Pwm (const Pwm &) = delete;
    Pwm &operator= (const Pwm &) = delete;
    bool ok () const { return status_ == 0; }
    int duty (int d) { return gk_pwm (gpio_, freq_, d); } // 0 .. 10000
    int frequency (int hz, int d) { freq_ = hz; return gk_pwm (gpio_, hz, d); }
    int servo (int pulse_us) { freq_ = 50; return gk_servo (gpio_, pulse_us); }
private:
    int gpio_, freq_, status_;
};

```

A device on the I2C bus (the bus opened by the first one made, if it was not).

```

class I2CDevice
{
public:
    explicit I2CDevice (int addr, int clock_hz = 0) : addr_ (addr) { status_ =
gk_i2c_open (clock_hz); }
    bool ok () const { return status_ == 0; }
    int addr () const { return addr_; }
    int write (const void *d, int n) const { return gk_i2c_write (addr_, d, n); }
    int read (void *d, int n) const { return gk_i2c_read (addr_, d, n); }
    int write_read (const void *w, int wn, void *r, int rn) const { return
gk_i2c_write_read (addr_, w, wn, r, rn); }
    int reg (int r) const { return gk_i2c_reg_read (addr_, r); }
    int reg (int r, int v) const { return gk_i2c_reg_write (addr_, r, v); }
private:
    int addr_, status_;
};

```